

Eigenvalues and inverse problems

Find a near-largest nonnormal eigenvalue using few matrix-vector products

Difficulty: challenging **Importance:** interesting to the community

Status: Open

Topic: nonsymmetric eigenvalue computation; query complexity

Rating rationale: Challenging reflects a query bound for general nonnormal matrices, whose eigenvalues need not be controlled by power iteration; community impact is the complexity of large nonsymmetric eigenproblems.

Context and notation

All norms are Euclidean vector norms or their induced matrix norms.

Problem statement

An unknown diagonalizable $A \in \mathbb{C}^{n \times n}$ is available only through exact queries $v \mapsto Av$. A number $K \geq 1$ is supplied, with the promise that $A = V\Lambda V^{-1}$ for some diagonal Λ and $\|V\|_2\|V^{-1}\|_2 \leq K$. Assume $\rho(A) = \max_{\mu \in \sigma(A)} |\mu| > 0$; neither the eigenvalues nor V are supplied.

Do universal constants $C, a, b > 0$ and a randomized algorithm exist that, for every such input and every $\varepsilon \in (0, 1/2)$, returns a complex number z with probability at least 0.99 such that

$$\text{there exists } \mu \in \sigma(A) : \quad |\mu| \geq (1 - \varepsilon)\rho(A), \quad |z - \mu| \leq \varepsilon\rho(A),$$

using at most $C[1 + \log(nK)]^a \varepsilon^{-b}$ matrix-vector queries? The algorithm may perform finite exact arithmetic between queries; the requested bound concerns query count. Products with A^* and solutions of shifted linear systems are not additional oracle operations. Supplying K and fixing a success probability make the computational formulation explicit.

References

Amsel et al., [workshop report](#), Problem 3.9. Shah, Srivastava, and Zeng, *Sparse Pseudospectral Shattering*, [first version](#), §1.3, Theorem 1.5 and its power-iteration argument, supplies the related spectral-radius estimate cited by the workshop.

Earlier status check — 2026-09-08

Searches for *nonnormal eigenvalue query complexity 2026* and *nonnormal spectral radius matrix-vector algorithm 2026* found no resolution. The radius estimate controls a nonnegative scalar and allows backward error; it does not locate a near-extremal eigenvalue in the complex plane as required here. The [April 2026 third version](#) replaces that application with a GMRES application (§6), so the version-specific locator above is intentional. The workshop still poses Problem 3.9 in its August update, separately from its resolved general Ritz-compression question.

Audit update — 2026-09-10

Rechecked [workshop version 3](#), Problem 3.9. Searches for nonnormal largest-eigenvalue matrix-vector query bounds found no theorem or lower bound settling the displayed question. Hermitian or singular-value oracle results do not answer it.